

מעבדה בתכנון ספרתי מתקדם

דו"ח מסכם 8

מגישים:

אלעד אסבג – 318530482

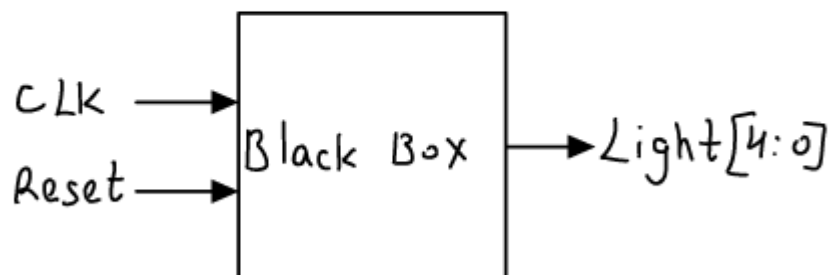
יוסי ברים- 315512848

צוות 7

B1:

SIGNALS	I/O	Discription
CLK	IN	תדר השעון
RESET	IN	Active hige syncronoce
LIGHT[4:0]	OUT	LIGHT[4:3] הולכי רגל- LIGHT[2:0] מכוניות-

Black Box



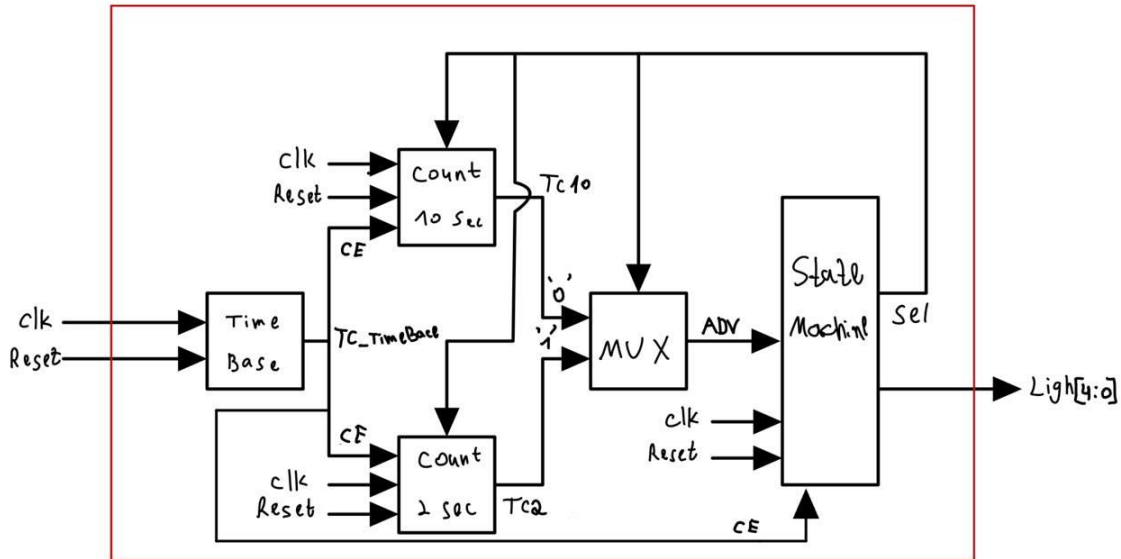
entity:

```
4
5 entity Top_level is
6 port(
7     CLK                : in  std_logic;
8     RESET               : in  std_logic;
9     LIGHT[4 downto 0]  : out  std_logic
10
11 );
12 end Top_level;
```

XDC:

```
2 #####
3 # Clocks #
4 #####
5 create_clock -name CLK -period 10 -waveform {0 5} [get_ports CLK]
6
7 #####
8 # Input Delay #
9 #####
10 set_input_delay 0 -clock CLK -add_delay [get_ports {RESET*}]
11
12 #####
13 # Output Delay #
14 #####
15 set_output_delay 0 -clock CLK -add_delay [get_ports {LIGHT*}]
16
17 #####
18 # Pin Locations and Voltages #
19 #####
20 set_property -dict { PACKAGE_PIN W5 IOSTANDARD LVCMOS33 } [get_ports { CLK }];
21 set_property -dict { PACKAGE_PIN U18 IOSTANDARD LVCMOS33 } [get_ports { RESET}];
22
23 set_property -dict { PACKAGE_PIN U16 IOSTANDARD LVCMOS33 } [get_ports { LIGHT[0] }];
24 set_property -dict { PACKAGE_PIN E19 IOSTANDARD LVCMOS33 } [get_ports { LIGHT[1] }];
25 set_property -dict { PACKAGE_PIN U19 IOSTANDARD LVCMOS33 } [get_ports { LIGHT[2] }];
26 set_property -dict { PACKAGE_PIN P1 IOSTANDARD LVCMOS33 } [get_ports { LIGHT[3] }];
27 set_property -dict { PACKAGE_PIN L1 IOSTANDARD LVCMOS33 } [get_ports { LIGHT[4] }];
28
29
30
31 #####
32 # Configuration Settings #
33 #####
34 set_property CFGBVS VCCO [current_design]
35 set_property CONFIG_VOLTAGE 3.3 [current_design]
```

B2:

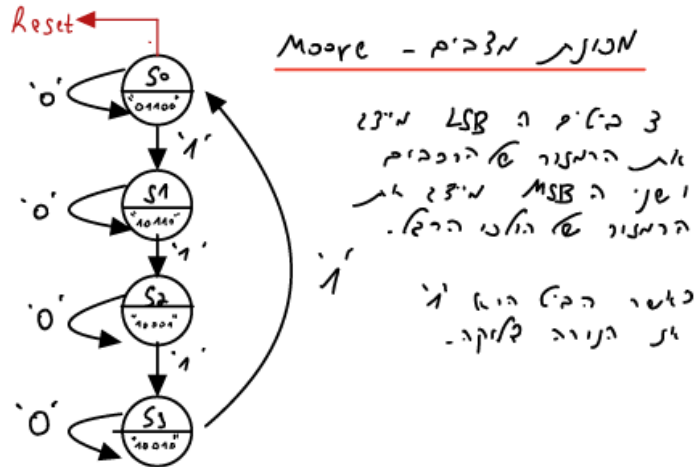


B3:

Entity:

```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.All;
3  use IEEE.STD_LOGIC_UNSIGNED.ALL;
4
5  entity State_machine is
6  port
7  (
8      CLK      :    in std_logic;
9      RESET    :    in std_logic;
10     TC       :    in std_logic;
11     SEL      :    out std_logic;
12     LIGHT    :    out std_logic_vector(4 downto 0)
13 );
14 end State_machine;
```

bubble diagram:



```

1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3  use IEEE.STD_LOGIC_UNSIGNED.ALL;
4
5  entity State_machine is
6  port
7  (
8  CLK      :    in std_logic;
9  RESET    :    in std_logic;
10 ADV      :    in std_logic;
11 CE       :    in std_logic;
12 SEL      :    out std_logic;
13 LIGHT    :    out std_logic_vector(4 downto 0)
14 );
15 end State_machine;
16
17 architecture State_machine_ARC of State_machine is
18     type StateType is (S0, S1, S2, S3);
19     signal STATE: StateType;
20 begin
21     p0: process
22     begin
23         wait until rising_edge(CLK);
24         if (RESET = '1') then
25             STATE <= S0;
26             LIGHT <= "01100";
27             SEL <= '0';
28         elsif (CE = '1') then
29             case STATE is
30                 when S0 =>
31                     if (ADV = '1') then
32                         STATE <= S1;
33                         LIGHT <= "10110";
34                         SEL <= '1';
35                     else
36                         STATE <= S0;
37                         LIGHT <= "01100";
38                         SEL <= '0';
39                     end if;

```

```

40
41
42     when S1 =>
43         if (ADV = '1') then
44             STATE <= S2;
45             LIGHT <= "10001";
46             SEL <= '0';
47         else
48             STATE <= S1;
49             LIGHT <= "10110";
50             SEL <= '1';
51         end if;
52
53     when S2 =>
54         if (ADV = '1') then
55             STATE <= S3;
56             LIGHT <= "10010";
57             SEL <= '1';
58         else
59             STATE <= S2;
60             LIGHT <= "10001";
61             SEL <= '0';
62         end if;
63
64     when S3 =>
65         if (ADV = '1') then
66             STATE <= S0;
67             LIGHT <= "01100";
68             SEL <= '0';
69         else
70             STATE <= S3;
71             LIGHT <= "10010";
72             SEL <= '1';
73         end if;
74     end case;
75 end if;
76 end process;
77 end State_machine_ARC;

```

```

1  library IEEE;
2  use IEEE.STD_LOGIC_1164.All;
3  use IEEE.STD_LOGIC_UNSIGNED.ALL;
4
5  entity Counters is
6  port(
7      CLK      : in std_logic;
8      RESET    : in std_logic;
9      CE       : in std_logic;
10     SEL      : in std_logic;
11     ADV      : out std_logic;
12 );
13 end Counters;
14
15 architecture Counters_ARC of Counters is
16     signal Q_2Seconds : std_logic_vector(1 downto 0);
17     signal Q_10Seconds : std_logic_vector(3 downto 0);
18     signal TC2, TC10 : std_logic;
19 begin
20     -- 2 Seconds --
21
22     Seconds2: process
23     begin
24         wait until rising_edge(CLK);
25         if (RESET = '1') then
26             Q_2Seconds <= (others => '0');
27             TC2 <= '0';
28         else
29             if (CE = '1') then
30                 if (SEL = '1') then
31                     if (Q_2Seconds < 1) then
32                         Q_2Seconds <= Q_2Seconds + 1;
33                         TC2 <= '0';
34                     else
35                         Q_2Seconds <= (others => '0');
36                         TC2 <= '1';
37                     end if;
38                 else
39                     Q_2Seconds <= (others => '0');
40                     TC2 <= '0';
41                 end if;
42             end if;
43         end if;
44     end process;
45

```

```

47     Seconds10: process
48     begin
49         wait until rising_edge(CLK);
50         if (RESET = '1') then
51             Q_10Seconds <= (others => '0');
52             TC10 <= '0';
53         else
54             if (CE = '1') then
55                 if (SEL = '0') then
56                     if (Q_10Seconds < 9) then
57                         Q_10Seconds <= Q_10Seconds + 1;
58                         TC10 <= '0';
59                     else
60                         Q_10Seconds <= (others => '0');
61                         TC10 <= '1';
62                     end if;
63                 else
64                     Q_10Seconds <= (others => '0');
65                     TC10 <= '0';
66                 end if;
67             end if;
68         end if;
69     end process;
70
71     -- ADVANCE MUX--
72
73     MUX: process (SEL, TC10, TC2)
74     begin
75         case SEL is
76             when '0' => ADV <= TC10;
77             when '1' => ADV <= TC2;
78             when others => ADV <= '0';
79         end case;
80     end process;
81
82 end Counters_ARC;
83
84

```

```

1  library IEEE;
2  use IEEE.STD_LOGIC_1164.All;
3  use IEEE.STD_LOGIC_UNSIGNED.ALL;
4
5  entity Time_Base is
6  port(
7      CLK      : in std_logic;
8      RESET    : in std_logic;
9      TC_TimeBase: out std_logic
10 );
11 end Time_Base;
12
13
14 architecture Time_Base_ARC of Time_Base is
15     signal Q : std_logic_VECTOR(26 downto 0);
16 begin
17     -----tc time base and mux for test mode
18
19     process
20     begin
21         wait until rising_edge(CLK);
22         if (RESET = '1') then
23             Q <= (others => '0');
24             TC_TimeBase <= '0';
25         else
26             if (Q < 99999999) then
27                 Q <= Q + 1;
28                 TC_TimeBase <= '0';
29             else
30                 Q <= (others => '0');
31                 TC_TimeBase <= '1';
32             end if;
33         end if;
34     end process;
35
36 end Time_Base_ARC;
37
38
39

```

```

1  library IEEE;
2  use IEEE.STD_LOGIC_1164.All;
3  use IEEE.STD_LOGIC_UNSIGNED.ALL;
4
5  entity Top_level is
6  port(
7      CLK          : in std_logic;
8      RESET        : in std_logic;
9      LIGHT        : out std_logic_VECTOR(4 downto 0)
10 );
11
12 end Top_level;
13
14 architecture ARC_Top_level of Top_level is
15
16 component State_machine
17 port(
18     CLK          : in std_logic;
19     RESET        : in std_logic;
20     ADV          : in std_logic;
21     CE           : in std_logic;
22     SEL          : out std_logic;
23     LIGHT        : out std_logic_vector(4 downto 0)
24 );
25
26 end component;
27
28 component Counters
29 port(
30     CLK          : in std_logic;
31     RESET        : in std_logic;
32     CE           : in std_logic;
33     SEL          : in std_logic;
34     ADV          : out std_logic
35 );
36
37 end component;
38
39 component Time_Base
40 port(
41     CLK          : in std_logic;
42     RESET        : in std_logic;
43     TC_TimeBase : out std_logic
44 );
45
46 end component;

```

```

45
46 signal SEL  : std_logic;
47 signal ADV  : std_logic;
48 signal CE   : std_logic;
49
50 begin
51     L0 : State_machine
52         port map(
53             CLK      => CLK,
54             RESET    => RESET,
55             ADV      => ADV,
56             CE       => CE,
57             SEL      => SEL,
58             LIGHT    => LIGHT
59         );
60
61     L1 : Counters
62         port map(
63             CLK      => CLK,
64             RESET    => RESET,
65             CE       => CE,
66             SEL      => SEL,
67             ADV      => ADV
68         );
69
70     L2 : Time_Base
71         port map(
72             CLK      => CLK,
73             RESET    => RESET,
74             TC_TimeBase => CE
75         );
76
77
78 end ARC_Top_level;

```

```

1  library IEEE;
2  use IEEE.STD_LOGIC_1164.All;
3  use IEEE.STD_LOGIC_UNSIGNED.ALL;
4
5  entity TB_Top_level is
6  end TB_Top_level;
7
8  architecture TB_ARC_Top_level of TB_Top_level is
9
10 component Top_level
11 port(
12     CLK          : in std_logic;
13     RESET        : in std_logic;
14     LIGHT        : out std_logic_VECTOR(4 downto 0)
15 );
16
17 end component;
18
19 signal TB_CLK          : std_logic:= '0' ;
20 signal TB_RESET        : std_logic:= '0' ;
21 signal TB_LIGHT        : std_logic_VECTOR(4 downto 0);
22
23
24 begin
25     DUT : Top_level
26     PORT map(
27
28     CLK      => TB_CLK,
29     RESET    => TB_RESET,
30     LIGHT    => TB_LIGHT
31 );
32
33
34 stim_proc: process
35 begin
36     wait for 5 ns;
37     TB_CLK <= not(TB_CLK);
38 end process;
39
40
41 TB_RESET <= '1',
42          '0' after 117 ns;
43
44 end TB_ARC_Top_level;

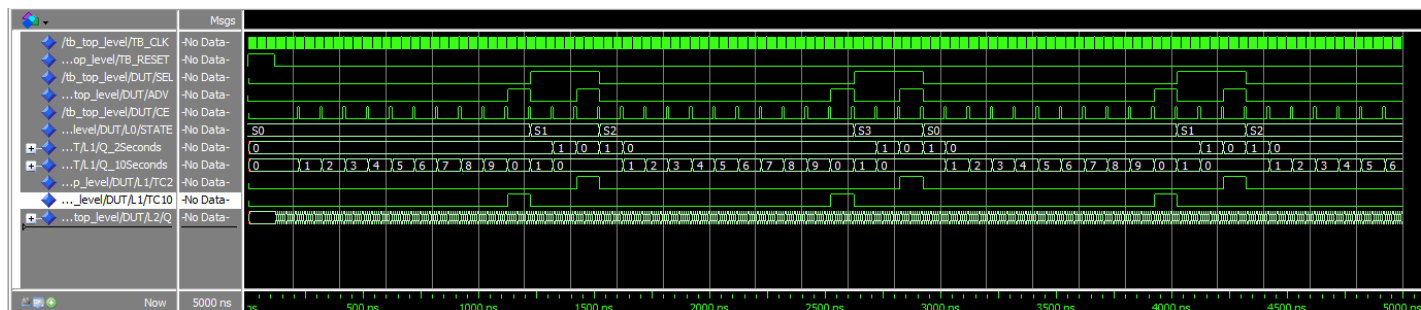
```

ATP:

תוכנית זו מיועדת למערכת רמזורים שנועדה לשלוט ב-4 מצבים שונים של 2 רמזורים מסונכרנים:

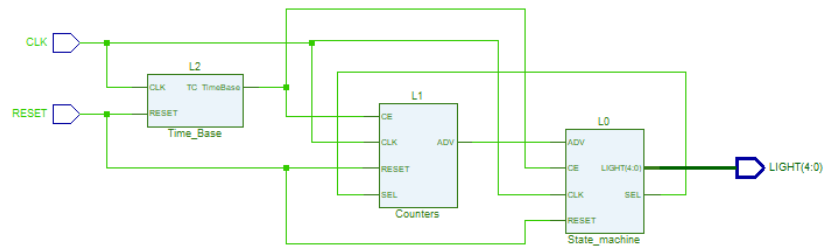
1. ירוק להולכי רגל ואדום לרכבים - למשך 10 שניות
2. אדום להולכי רגל ואדום צהוב לרכבים - למשך 2 שניות
3. אדום להולכי רגל וירוק לרכבים - למשך 10 שניות
4. אדום להולכי רגל וצהוב לרכבים - למשך 2 שניות

תדר הבורד הוא 100 מגה הרץ, נצטרך להמיר תדר זה לשניות באמצעות:
Time Base
אנו נוודא שבכל מצב 2 הרמזורים מסונכרנים ושהאורות מופעלים למשך הזמן הרצוי.

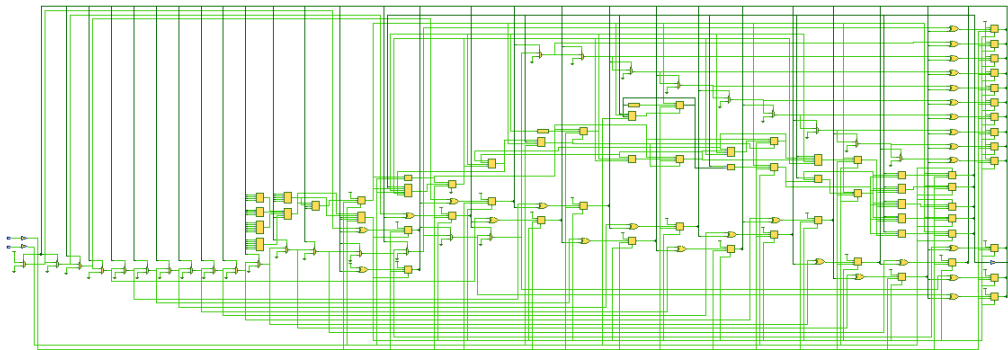


Synthesis:

RTL:



TEC:



AREA:

```
// Precision RTL Plus Validate 64-bit 2021.1.0.4 (Production Release) Tue Jul 20 01:22:31 PDT 2021
//
// Copyright (c) Mentor Graphics Corporation, 1996-2021, All Rights Reserved.
// Portions copyright 1991-2008 Compuware Corporation
// UNPUBLISHED, LICENSED SOFTWARE.
// CONFIDENTIAL AND PROPRIETARY INFORMATION WHICH IS THE
// PROPERTY OF MENTOR GRAPHICS CORPORATION OR ITS LICENSORS
//
// Running on Windows 10 eladjose@ELEC-403-103 10.0.19041 x64
//
// Start time Wed Aug 06 18:29:12 2025

*****
Device Utilization for 7A35TCPG236
*****
Resource          Used   Avail   Utilization
-----
IOs                7      106     6.60%
Global Buffers     1       32     3.13%
LUTs              17     20900    0.08%
CLB Slices         6      8150    0.07%
Dffs or Latches    41     41600    0.10%
Block RAMs         0       50     0.00%
DSP48E1s          0       90     0.00%
-----

*****

Library: work      Cell: Top_level      View: ARC_Top_level

*****

Cell      Library References      Total Area

BUFGP      xca7      1 x
FDRE      xca7      39 x      1      39 Dffs or Latches
FDSE      xca7      2 x      1      2 Dffs or Latches
GND      xca7      1 x
IBUF      xca7      1 x
INV      xca7      2 x      1      2 LUTs
LUT2      xca7      2 x      1      2 LUTs
LUT3      xca7      4 x      1      4 LUTs
LUT4      xca7     10 x      1     10 LUTs
LUT5      xca7      4 x      1      4 LUTs
LUT6      xca7      3 x      1      3 LUTs
MUXCY      xca7     26 x      1     26 MUX CARRYs
OBUF      xca7      5 x
VCC      xca7      1 x
XORCY      xca7     27 x

Number of ports :          7
Number of nets :         130
Number of instances :       128
Number of references to this view :      0

Total accumulated area :
Number of Dffs or Latches :          41
Number of LUTs :          17
Number of Primitive LUTs :          25
Number of LUTs with LUTNM/HLUTNM :          16
Number of MUX CARRYs :          26
Number of accumulated instances :       128

*****
IO Register Mapping Report
*****
Design: work.Top_level.ARC_Top_level

+-----+-----+-----+-----+-----+
| Port | Direction | INFF | OUTFF | TRIFF |
+-----+-----+-----+-----+-----+
| CLK | Input | | | |
+-----+-----+-----+-----+-----+
| RESET | Input | | | |
+-----+-----+-----+-----+-----+
| LIGHT(4) | Output | | | |
+-----+-----+-----+-----+-----+
| LIGHT(3) | Output | | | |
+-----+-----+-----+-----+-----+
| LIGHT(2) | Output | | | |
+-----+-----+-----+-----+-----+
| LIGHT(1) | Output | | | |
+-----+-----+-----+-----+-----+
| LIGHT(0) | Output | | | |
+-----+-----+-----+-----+-----+
Total registers mapped: 0
```

TIMING:

```
1 // Precision RTL Plus Validate 64-bit 2021.1.0.4 (Production Release) Tue Jul 20 01:22:31 PDT 2021
2 //
3 // Copyright (c) Mentor Graphics Corporation, 1996-2021. All Rights Reserved.
4 // Portions copyright 1991-2008 Compware Corporation
5 // UNPUBLISHED, LICENSED SOFTWARE.
6 // CONFIDENTIAL AND PROPRIETARY INFORMATION WHICH IS THE
7 // PROPERTY OF MENTOR GRAPHICS CORPORATION OR ITS LICENSORS
8 //
9 // Running on Windows 10 eladjose@ELDC-403-103 10.0.19041 x64
10 //
11 // Start time Wed Aug 06 18:29:12 2025
12 //
13 -- Device: Xilinx - ARTIX-7 : 7AS10TCFG236 : 1
14 -- CTE reports summary..
15 -- POST-SYNTHESIS TIMING REPORTS ARE ESTIMATES AND SHOULD NOT BE RELIED ON TO MAKE QoR DECISIONS.
16 For accurate timing information, please run place-and-route (PAR) and review PAR generated timing reports.
17
18 =====
19 Clock Frequency Report
20 =====
21
22 Domain Clock Name Min Period (Freq) Required Period (Freq)
23 -----
24 CLK_PS CLK 6.041 (165.536 MHz) 10.000 (100.000 MHz)
25 CLK_PS VirtualClock 6.041 (165.536 MHz) 10.000 (100.000 MHz)
26
27 =====
28 Setup Timing Analysis of VirtualClock
29 =====
30 Setup Slack Path Summary
31 =====
32
33 Index Setup Path Source Data Start Pin Data End Edge Logic
34 -----
35 1 3.959 3.959 CLK VirtualClock L0_reg_STATE(0)/C LIGHT(1) Rise 1
36 2 4.192 3.720 CLK VirtualClock L0_reg_LIGHT(4)/C LIGHT(4) Rise 1
37 3 4.192 3.720 CLK VirtualClock L0_reg_LIGHT(0)/C LIGHT(0) Rise 1
38 4 4.172 3.650 CLK VirtualClock L0_reg_LIGHT(3)/C LIGHT(3) Rise 1
39 5 4.172 3.650 CLK VirtualClock L0_reg_LIGHT(2)/C LIGHT(2) Rise 1
40
41 Showing 5 paths found in the design.
42 CTE Path Report
43
44 Critical path #1, (path slack = 3.959): Logic Levels = 1
45
46 SOURCE CLOCK: name: CLK period: 10.000000
47 Times are relative to the 1st rising edge
48 DEST CLOCK: name: VirtualClock period: 10.000000
49 Times are relative to the 2nd rising edge
50
51 NAME GATE DELAY ARRIVAL DIR FANOUT LEVEL
52 -----
53 L0_reg_STATE(0)/C FBRE 0.000 up
54 L0_reg_STATE(0)/Q FBRE 0.496 0.496 up 12 0
55 SET (net) 0.506
56 LIGHT_obuf(1)/I OBUF 1.002 up
57 LIGHT_obuf(1)/O OBUF 2.991 3.993 up 0 1
58 LIGHT(1) (net) 0.000
59 LIGHT(1) (port) 2.993 up
60
61 Initial edge separation: 10.000
62 Source clock delay: - 2.149
63 Dest clock delay: + 0.000
64
65 Edge separation: -----
66 Setup constraints: - 0.000
67
68 Data required time: 7.852
69 Data arrival time: - 3.893 ( 87.00% cell delay, 13.00% net delay )
70
71 Slack: 3.959
72
73 =====
74 Setup Timing Analysis of CLK
75 =====
76 Setup Slack Path Summary
77 =====
78
79 Index Setup Path Source Data Start Pin Data End Edge Logic
80 -----
81 1 6.906 2.694 CLK CLK reg_q(17)/C reg_q(26)/R Rise 3
82 2 6.919 2.691 CLK CLK reg_q(10)/C reg_q(26)/R Rise 3
83 3 6.919 2.691 CLK CLK reg_q(18)/C reg_q(26)/R Rise 3
84 4 6.919 2.691 CLK CLK reg_q(5)/C reg_q(26)/R Rise 3
85 5 6.919 2.691 CLK CLK reg_q(14)/C reg_q(26)/R Rise 3
86 6 6.919 2.691 CLK CLK reg_q(9)/C reg_q(26)/R Rise 3
87 7 6.919 2.691 CLK CLK reg_q(17)/C reg_q(26)/R Rise 3
88 8 6.919 2.691 CLK CLK reg_q(12)/C reg_q(26)/R Rise 3
89 9 6.919 2.691 CLK CLK reg_q(2)_L_1/C reg_q(26)/R Rise 3
90 10 6.919 2.691 CLK CLK reg_q(3)_L_0/C reg_q(26)/R Rise 3
91
92 CTE Path Report
93
94 Critical path #1, (path slack = 6.906): Logic Levels = 3
95
96 SOURCE CLOCK: name: CLK period: 10.000000
97 Times are relative to the 1st rising edge
98 DEST CLOCK: name: CLK period: 10.000000
99 Times are relative to the 2nd rising edge
100
101 NAME GATE DELAY ARRIVAL DIR FANOUT LEVEL
102 -----
103 reg_q(17)/C FBRE 0.496 0.000 up
104 reg_q(17)/Q FBRE 0.496 0.496 up 4 0
105 L0_Q(17) (net) 0.894
106 in38910s36271/I LUT6 0.124 0.850 up 1 1
107 in38910s36271/O LUT6 0.124 0.974 up 1 1
108 in38910s36271/O LUT6 0.124 1.307 up 1 1
109 in38910s36271/O LUT6 0.124 1.431 dm 2 2
110 in38910s36271/O LUT6 0.124 1.766 dm 2 2
111 in38910s36271/O LUT6 0.124 1.890 up 27 3
112 in38910s36271/O LUT6 0.804
113 reg_q(26)/R FBRE 2.694 up
114
115 Initial edge separation: 10.000
116 Source clock delay: - 2.149
117 Dest clock delay: + 2.149
118
119 Edge separation: -----
120 Setup constraints: - 0.400
121
122 Data required time: 9.600
123 Data arrival time: - 2.694 ( 32.24% cell delay, 67.76% net delay )
124
125 Slack: 6.906
126
127 =====
128 Input Delay Report
129 =====
130
131 Input Clock Name Slack (ns)
132 -----
133 RESET VirtualClock -----
134
135 =====
136 Output Delay Report
137 =====
138
139 Output Clock Name Slack (ns)
140 -----
141 LIGHT(1) VirtualClock 3.959
142 LIGHT(4) VirtualClock 4.192
143 LIGHT(0) VirtualClock 4.192
144 LIGHT(3) VirtualClock 4.172
145 LIGHT(2) VirtualClock 4.172
146
```

SYNTHESIS CODE THET REPORTS ABOUT THE STATE

MACHIN ENCODING:

```
# Info: [45144]: FSM: State encoding table.
# Info: [40000]: FSM:      Index      State Name      Literal      Encoding
# Info: [40000]: FSM:      0          S0             00           00
# Info: [40000]: FSM:      1          S1             01           01
# Info: [40000]: FSM:      2          S2             10           10
# Info: [40000]: FSM:      3          S3             11           11
# Info: [44506]: Module work.Counters(Counters_ARC): Pre-processing...
```

P AND R :

TIMING:

Copyright 1986-2020 Xilinx, Inc. All Rights Reserved.

| Tool Version : Vivado v.2020.1 (win64) Build 2902540 Wed May 27 19:54:49 MDT 2020

| Date : Wed Aug 6 18:57:05 2025

| Host : ELEC-403-103 running 64-bit major release (build 9200)

| Command : report_timing_summary -delay_type min_max -report_unconstrained -check_timing_verbose -max_paths 10 -input_pins -routable_nets -datasheet -name timing_3 -file Z:/VHDL2/lab_28/project_1/timing_report.txt

| Design : Top_Level

| Device : 7a35t-cpg236

| Speed File : -1 PRODUCTION 1.23 2018-06-13

Timing Summary Report

| Timer Settings

| -----

Enable Multi Corner Analysis : Yes

Enable Pessimism Removal : Yes

Pessimism Removal Resolution : Nearest Common Node

Enable Input Delay Default Clock : No

Enable Preset / Clear Arcs : No

Disable Flight Delays : No

Ignore I/O Paths : No

Timing Early Launch at Borrowing Latches : No

Borrow Time for Max Delay Exceptions : Yes

Merge Timing Exceptions : Yes

Corner Analyze Analyze

Name Max Paths Min Paths

Slow Yes Yes

Fast Yes Yes

check_timing report

Table of Contents

- 1. checking no_clock (0)
- 2. checking constant_clock (0)
- 3. checking pulse_width_clock (0)
- 4. checking unconstrained_internal_endpoints (0)
- 5. checking no_input_delay (0)
- 6. checking no_output_delay (0)
- 7. checking multiple_clock (0)
- 8. checking generated_clocks (0)
- 9. checking loops (0)
- 10. checking partial_input_delay (0)
- 11. checking partial_output_delay (0)
- 12. checking latch_loops (0)

1. checking no_clock (0)

There are 0 register/latch pins with no clock.

2. checking constant_clock (0)

There are 0 register/latch pins with constant_clock.

3. checking pulse_width_clock (0)

There are 0 register/latch pins which need pulse_width check

4. checking unconstrained_internal_endpoints (0)

There are 0 pins that are not constrained for maximum delay.

There are 0 pins that are not constrained for maximum delay due to constant clock.

5. checking no_input_delay (0)

There are 0 input ports with no input delay specified.

There are 0 input ports with no input delay but user has a false path constraint.

6. checking no_output_delay (0)

There are 0 ports with no output delay specified.

There are 0 ports with no output delay but user has a false path constraint

There are 0 ports with no output delay but with a timing clock defined on it or propagating through it

7. checking multiple_clock (0)

There are 0 register/latch pins with multiple clocks.

8. checking generated_clocks (0)

There are 0 generated clocks that are not connected to a clock source.

9. checking loops (0)

There are 0 combinational loops in the design.

10. checking partial_input_delay (0)

There are 0 input ports with partial input delay specified.

11. checking partial_output_delay (0)

There are 0 ports with partial output delay specified.

12. checking latch_loops (0)

There are 0 combinational latch loops in the design through latch input

| Design Timing Summary

WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	THS Total Endpoints	WPWS(ns)	TPWS(ns)	TPWS Failing Endpoints	TPWS Total Endpoints
-0.713	-3.424	5	92	0.191	0.000	0	92	4.500	0.000	0	45

Timing constraints are not met.

| Clock Summary

Clock	Waveform(ns)	Period(ns)	Frequency(MHz)
CLK	{0.000 5.000}	10.000	100.000

| Intra Clock Table

Clock Endpoints	WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	THS Total Endpoints	WPWS(ns)	TPWS(ns)	TPWS Failing Endpoints	TPWS Total Endpoints
CLK	-0.713	-3.424	5	92	0.191	0.000	0	92	4.500	0.000	0	45

| Inter Clock Table

From Clock	To Clock	WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	THS Total Endpoints
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

| Other Path Groups Table

| -----

Path Group	From Clock	To Clock	WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	THS Total Endpoints
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

| User Ignored Path Table

| -----

Path Group From Clock To Clock

----- ----- -----

| Unconstrained Path Table

| -----

Path Group From Clock To Clock

----- ----- -----

| Timing Details

| -----

From Clock: CLK

To Clock: CLK

Setup :	5 Failing Endpoints, Worst Slack	-0.713ns, Total Violation	-3.424ns
Hold :	0 Failing Endpoints, Worst Slack	0.191ns, Total Violation	0.000ns
PW :	0 Failing Endpoints, Worst Slack	4.500ns, Total Violation	0.000ns

Max Delay Paths

Slack (VIOLATED) : -0.713ns (required time - arrival time)

Source: L0/LIGHT_reg[1]_replica/C
(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: LIGHT[1]
(output port clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Max at Slow Process Corner

Requirement: 10.000ns (CLK rise@10.000ns - CLK rise@0.000ns)

Data Path Delay: 5.519ns (logic 3.986ns (72.220%) route 1.533ns (27.780%))

Logic Levels: 1 (OBUF=1)

Output Delay: 0.000ns

Clock Path Skew: -5.159ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): 0.000ns = (10.000 - 10.000)

Source Clock Delay (SCD): 5.159ns

Clock Pessimism Removal (CPR): 0.000ns

Clock Uncertainty: 0.035ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE

Total System Jitter (TSJ): 0.071ns

Total Input Jitter (TIJ): 0.000ns

Discrete Jitter (DJ): 0.000ns

Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	1.458	1.458	r CLK_IBUF_inst/O
	net (fo=1, routed)	1.967	3.425	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUF_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.096	3.521	r CLK_IBUF_BUF_inst/O
	net (fo=44, routed)	1.638	5.159	L0/CLK_IBUF_BUF
SLICE_X0Y43	FDRE			r L0/LIGHT_reg[1]_replica/C
SLICE_X0Y43	FDRE (Prop_fdre_C_Q)	0.456	5.615	r L0/LIGHT_reg[1]_replica/Q
	net (fo=1, routed)	1.533	7.148	LIGHT_OBUF[1]_repN_alias
E19				r LIGHT_OBUF[1]_inst/I
E19	OBUF (Prop_obuf_I_O)	3.530	10.678	r LIGHT_OBUF[1]_inst/O
	net (fo=0)	0.000	10.678	LIGHT[1]
E19				r LIGHT[1] (OUT)

(clock CLK rise edge)	10.000	10.000	r
clock pessimism	0.000	10.000	
clock uncertainty	-0.035	9.965	
output delay	-0.000	9.965	

required time	9.965	
arrival time	-10.678	

slack	-0.713	
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Slack (VIOLATED) : -0.685ns (required time - arrival time)

Source: L0/LIGHT_reg[0]/C

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: LIGHT[0]

(output port clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Max at Slow Process Corner

Requirement: 10.000ns (CLK rise@10.000ns - CLK rise@0.000ns)

Data Path Delay: 5.490ns (logic 3.961ns (72.141%) route 1.530ns (27.859%))

Logic Levels: 1 (OBUF=1)

Output Delay: 0.000ns

Clock Path Skew: -5.159ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): 0.000ns = (10.000 - 10.000)

Source Clock Delay (SCD): 5.159ns

Clock Pessimism Removal (CPR): 0.000ns

Clock Uncertainty: 0.035ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE

Total System Jitter (TSJ): 0.071ns

Total Input Jitter (TIJ): 0.000ns

Discrete Jitter (DJ): 0.000ns

Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5			r	CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_i_O)	1.458	1.458	r CLK_IBUF_inst/O
	net (fo=1, routed)	1.967	3.425	CLK_IBUF
BUFGCTRL_X0Y0			r	CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_i_O)	0.096	3.521	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	1.638	5.159	L0/CLK_IBUF_BUFG
SLICE_X1Y3	FDRE		r	L0/LIGHT_reg[0]/C

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output delay	-0.000	9.965
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arrival time -10.650

slack	-0.685
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(output port clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})
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Phase Error (PE): 0.000ns

W5	r CLK_IBUF_inst/I
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W5	IBUF (Prop_ibuf_i_O)	1.458	1.458	r	CLK_IBUF_inst/O
	net (fo=1, routed)	1.967	3.425		CLK_IBUF
BUFGCTRL_X0Y0				r	CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_i_O)	0.096	3.521	r	CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	1.622	5.143		L0/CLK_IBUF_BUFG
SLICE_X65Y87	FDRE			r	L0/LIGHT_reg[4]/C

SLICE_X65Y87	FDRE (Prop_fdre_C_Q)	0.456	5.599	r	L0/LIGHT_reg[4]/Q
	net (fo=1, routed)	1.526	7.124		LIGHT_OBUF[4]
L1				r	LIGHT_OBUF[4]_inst/I
L1	OBUF (Prop_obuf_i_O)	3.521	10.646	r	LIGHT_OBUF[4]_inst/O
	net (fo=0)	0.000	10.646		LIGHT[4]
L1				r	LIGHT[4] (OUT)

	(clock CLK rise edge)	10.000	10.000	r	
	clock pessimism	0.000	10.000		
	clock uncertainty	-0.035	9.965		
	output delay	-0.000	9.965		

	required time		9.965		
	arrival time		-10.646		

	slack		-0.681		
Slack (VIOLATED) : -0.676ns (required time - arrival time)					
Source:	L0/LIGHT_reg[3]/C				
	(rising edge-triggered cell FDSE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})				
Destination:	LIGHT[3]				
	(output port clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})				
Path Group:	CLK				
Path Type:	Max at Slow Process Corner				
Requirement:	10.000ns (CLK rise@10.000ns - CLK rise@0.000ns)				
Data Path Delay:	5.497ns (logic 3.971ns (72.248%) route 1.526ns (27.752%))				
Logic Levels:	1 (OBUF=1)				
Output Delay:	0.000ns				
Clock Path Skew:	-5.144ns (DCD - SCD + CPR)				
Destination Clock Delay (DCD):	0.000ns = (10.000 - 10.000)				
Source Clock Delay (SCD):	5.144ns				
Clock Pessimism Removal (CPR):	0.000ns				
Clock Uncertainty:	0.035ns ((TSJ^2 + TJU^2)^1/2 + DJ) / 2 + PE				
Total System Jitter (TSJ):	0.071ns				
Total Input Jitter (TJU):	0.000ns				
Discrete Jitter (DJ):	0.000ns				
Phase Error (PE):	0.000ns				

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_i_O)	1.458	1.458	r CLK_IBUF_inst/O
	net (fo=1, routed)	1.967	3.425	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_i_O)	0.096	3.521	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	1.623	5.144	L0/CLK_IBUF_BUFG
SLICE_X65Y61	FDSE			r L0/LIGHT_reg[3]/C

SLICE_X65Y61	FDSE (Prop_fdse_C_Q)	0.456	5.600	r L0/LIGHT_reg[3]/Q
	net (fo=1, routed)	1.526	7.125	LIGHT_OBUF[3]
P1				r LIGHT_OBUF[3]_inst/I
P1	OBUF (Prop_obuf_i_O)	3.515	10.641	r LIGHT_OBUF[3]_inst/O
	net (fo=0)	0.000	10.641	LIGHT[3]
P1				r LIGHT[3] (OUT)

	(clock CLK rise edge)	10.000	10.000	r
	clock pessimism	0.000	10.000	
	clock uncertainty	-0.035	9.965	
	output delay	-0.000	9.965	

	required time	9.965	
	arrival time	-10.641	

slack	-0.676
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Slack (VIOLATED) : -0.668ns (required time - arrival time)

Source: L0/LIGHT_reg[2]/C

(rising edge-triggered cell FDSE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: LIGHT[2]

(output port clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Max at Slow Process Corner

Requirement: 10.000ns (CLK rise@10.000ns - CLK rise@0.000ns)

Data Path Delay: 5.486ns (logic 3.957ns (72.122%) route 1.530ns (27.878%))

Logic Levels: 1 (OBUF=1)

Output Delay: 0.000ns

Clock Path Skew: -5.146ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): 0.000ns = (10.000 - 10.000)

Source Clock Delay (SCD): 5.146ns

Clock Pessimism Removal (CPR): 0.000ns

Clock Uncertainty: 0.035ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE

Total System Jitter (TSJ): 0.071ns

Total Input Jitter (TIJ): 0.000ns

Discrete Jitter (DJ): 0.000ns

Phase Error (PE): 0.000ns

Location Delay type Incr(ns) Path(ns) Netlist Resource(s)

(clock CLK rise edge) 0.000 0.000 r

W5 0.000 0.000 r CLK (IN)

net (fo=0) 0.000 0.000 CLK

W5 r CLK_IBUF_inst/I

W5 IBUF (Prop_ibuf_I_O) 1.458 1.458 r CLK_IBUF_inst/O

net (fo=1, routed) 1.967 3.425 CLK_IBUF

BUFGCTRL_X0Y0 r CLK_IBUF_BUFG_inst/I

BUFGCTRL_X0Y0 BUFG (Prop_bufg_I_O) 0.096 3.521 r CLK_IBUF_BUFG_inst/O

net (fo=44, routed) 1.625 5.146 L0/CLK_IBUF_BUFG

SLICE_X1Y20 FDSE r L0/LIGHT_reg[2]/C

SLICE_X1Y20 FDSE (Prop_fdse_C_Q) 0.456 5.602 r L0/LIGHT_reg[2]/Q

net (fo=1, routed) 1.530 7.132 LIGHT_OBUF[2]

U19 r LIGHT_OBUF[2]_inst/I

U19 OBUF (Prop_obuf_I_O) 3.501 10.633 r LIGHT_OBUF[2]_inst/O

net (fo=0) 0.000 10.633 LIGHT[2]

U19 r LIGHT[2] (OUT)

(clock CLK rise edge) 10.000 10.000 r

clock pessimism 0.000 10.000

clock uncertainty -0.035 9.965

output delay -0.000 9.965

required time 9.965

arrival time -10.633

slack -0.668

Slack (MET) : 3.419ns (required time - arrival time)

Source: RESET

(input port clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: L0/LIGHT_reg[4]/R

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Setup (Max at Slow Process Corner)

Requirement: 10.000ns (CLK rise@10.000ns - CLK rise@0.000ns)

Data Path Delay: 10.962ns (logic 1.441ns (13.148%) route 9.521ns (86.852%))

Logic Levels: 1 (IBUF=1)

Input Delay: 0.000ns

Clock Path Skew: 4.846ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): 4.846ns = (14.846 - 10.000)

Source Clock Delay (SCD): 0.000ns

Clock Pessimism Removal (CPR): 0.000ns

Clock Uncertainty: 0.035ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE

Total System Jitter (TSJ): 0.071ns

Total Input Jitter (TIJ): 0.000ns

Discrete Jitter (DJ): 0.000ns

Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)

	(clock CLK rise edge)	0.000	0.000	r
	input delay	0.000	0.000	
U18		0.000	0.000	r RESET (IN)
	net (fo=0)	0.000	0.000	RESET
U18				r RESET_IBUF_inst/I
U18	IBUF (Prop_ibuf_i_O)	1.441	1.441	r RESET_IBUF_inst/O
	net (fo=14, routed)	9.521	10.962	L0/RESET_IBUF
SLICE_X6SY87	FDRE			r L0/LIGHT_reg[4]/R

	(clock CLK rise edge)	10.000	10.000	r
W5		0.000	10.000	r CLK (IN)
	net (fo=0)	0.000	10.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_i_O)	1.388	11.388	r CLK_IBUF_inst/O
	net (fo=1, routed)	1.862	13.250	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_i_O)	0.091	13.341	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	1.505	14.846	L0/CLK_IBUF_BUFG
SLICE_X6SY87	FDRE			r L0/LIGHT_reg[4]/C
	clock pessimism	0.000	14.846	
	clock uncertainty	-0.035	14.810	
SLICE_X6SY87	FDRE (Setup_fdre_C_R)	-0.429	14.381	L0/LIGHT_reg[4]

	required time		14.381	
	arrival time		-10.962	

	slack		3.419	

Slack (MET) : 4.209ns (required time - arrival time)

Source: RESET

(input port clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: L2/Q_reg[4]/R

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Setup (Max at Slow Process Corner)

Requirement: 10.000ns (CLK rise@10.000ns - CLK rise@0.000ns)

Data Path Delay: 10.019ns (logic 1.565ns (15.624%) route 8.453ns (84.376%))

Logic Levels: 2 (IBUF=1 LUT2=1)

Input Delay: 0.000ns

Clock Path Skew: 4.786ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): 4.786ns = (14.786 - 10.000)

Source Clock Delay (SCD): 0.000ns

Clock Pessimism Removal (CPR): 0.000ns

Clock Uncertainty: 0.035ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE

Total System Jitter (TSJ): 0.071ns

Total Input Jitter (TIJ): 0.000ns

Discrete Jitter (DJ): 0.000ns

Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
	input delay	0.000	0.000	
U18		0.000	0.000	r RESET (IN)
	net (fo=0)	0.000	0.000	RESET
U18			r	RESET_IBUF_inst/I
U18	IBUF (Prop_ibuf_I_O)	1.441	1.441	r RESET_IBUF_inst/O
	net (fo=14, routed)	6.948	8.389	L2/RESET_IBUF
SLICE_X28Y45			r	L2/Q[0]_i_1/O
SLICE_X28Y45	LUT2 (Prop_lut2_I_O_0)	0.124	8.513	r L2/Q[0]_i_1/O
	net (fo=27, routed)	1.505	10.019	L2/Q[0]_i_1_n_0
SLICE_X30Y43	FDRE		r	L2/Q_reg[4]/R
	(clock CLK rise edge)	10.000	10.000	r
W5		0.000	10.000	r CLK (IN)
	net (fo=0)	0.000	10.000	CLK
W5			r	CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	1.388	11.388	r CLK_IBUF_inst/O
	net (fo=1, routed)	1.862	13.250	CLK_IBUF
BUFGCTRL_X0Y0			r	CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.091	13.341	r CLK_IBUF_BUFG_inst/O

net (fo=44, routed) 1.445 14.786 L2/CLK_IBUF_BUFG

SLICE_X30Y43 FDRE r L2/Q_reg[4]/C

clock pessimism 0.000 14.786

clock uncertainty -0.035 14.751

SLICE_X30Y43 FDRE (Setup_fdre_C_R) -0.524 14.227 L2/Q_reg[4]

required time 14.227

arrival time -10.019

slack 4.209

Slack (MET) : 4.209ns (required time - arrival time)

Source: RESET

(input port clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: L2/Q_reg[5]/R

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Setup (Max at Slow Process Corner)

Requirement: 10.000ns (CLK rise@10.000ns - CLK rise@0.000ns)

Data Path Delay: 10.019ns (logic 1.565ns (15.624%) route 8.453ns (84.376%))

Logic Levels: 2 (IBUF=1 LUT2=1)

Input Delay: 0.000ns

Clock Path Skew: 4.786ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): 4.786ns = (14.786 - 10.000)

Source Clock Delay (SCD): 0.000ns

Clock Pessimism Removal (CPR): 0.000ns

Clock Uncertainty: 0.035ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE

Total System Jitter (TSJ): 0.071ns

Total Input Jitter (TIJ): 0.000ns

Discrete Jitter (DJ): 0.000ns

Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
	input delay	0.000	0.000	
U18		0.000	0.000	r RESET (IN)
	net (fo=0)	0.000	0.000	RESET
U18				r RESET_IBUF_inst/I
U18	IBUF (Prop_ibuf_I_O)	1.441	1.441	r RESET_IBUF_inst/O
	net (fo=14, routed)	6.948	8.389	L2/RESET_IBUF
SLICE_X28Y45				r L2/Q[0]_i_1/I0
SLICE_X28Y45	LUT2 (Prop_lut2_I0_O)	0.124	8.513	r L2/Q[0]_i_1/O
	net (fo=27, routed)	1.505	10.019	L2/Q[0]_i_1_n_0
SLICE_X30Y43	FDRE			r L2/Q_reg[5]/R

	(clock CLK rise edge)	10.000	10.000	r
W5		0.000	10.000	r CLK (IN)
	net (fo=0)	0.000	10.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	1.388	11.388	r CLK_IBUF_inst/O
	net (fo=1, routed)	1.862	13.250	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.091	13.341	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	1.445	14.786	L2/CLK_IBUF_BUFG
SLICE_X30Y43	FDRE			r L2/Q_reg[5]/C
	clock pessimism	0.000	14.786	
	clock uncertainty	-0.035	14.751	
SLICE_X30Y43	FDRE (Setup_fdre_C_R)	-0.524	14.227	L2/Q_reg[5]

	required time		14.227	
	arrival time		-10.019	

	slack		4.209	
Slack (MET) : 4.209ns (required time - arrival time)				
Source:	RESET			
	(input port clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})			
Destination:	L2/Q_reg[6]/R			
	(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})			
Path Group:	CLK			
Path Type:	Setup (Max at Slow Process Corner)			
Requirement:	10.000ns (CLK rise@10.000ns - CLK rise@0.000ns)			
Data Path Delay:	10.019ns (logic 1.565ns (15.624%) route 8.453ns (84.376%))			
Logic Levels:	2 (IBUF=1 LUT2=1)			
Input Delay:	0.000ns			
Clock Path Skew:	4.786ns (DCD - SCD + CPR)			
Destination Clock Delay (DCD):	4.786ns = (14.786 - 10.000)			
Source Clock Delay (SCD):	0.000ns			
Clock Pessimism Removal (CPR):	0.000ns			
Clock Uncertainty:	0.035ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE			
Total System Jitter (TSJ):	0.071ns			
Total Input Jitter (TIJ):	0.000ns			
Discrete Jitter (DJ):	0.000ns			
Phase Error (PE):	0.000ns			

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)

	(clock CLK rise edge)	0.000	0.000	r

```

input delay      0.000  0.000

U18              0.000  0.000 r RESET (IN)

net (fo=0)       0.000  0.000  RESET

U18              r RESET_IBUF_inst/I

U18      IBUF (Prop_ibuf_I_O)  1.441  1.441 r RESET_IBUF_inst/O

net (fo=14, routed)  6.948  8.389  L2/RESET_IBUF

SLICE_X28Y45     r L2/Q[0]_i_1/I0

SLICE_X28Y45     LUT2 (Prop_lut2_I0_O)  0.124  8.513 r L2/Q[0]_i_1/O

net (fo=27, routed)  1.505  10.019  L2/Q[0]_i_1_n_0

SLICE_X30Y43     FDRE              r L2/Q_reg[6]/R
-----
```

```

(clock CLK rise edge)  10.000  10.000 r

W5              0.000  10.000 r CLK (IN)

net (fo=0)       0.000  10.000  CLK

W5              r CLK_IBUF_inst/I

W5      IBUF (Prop_ibuf_I_O)  1.388  11.388 r CLK_IBUF_inst/O

net (fo=1, routed)  1.862  13.250  CLK_IBUF

BUFCTRL_X0Y0     r CLK_IBUF_BUF_inst/I

BUFCTRL_X0Y0     BUFG (Prop_bufg_I_O)  0.091  13.341 r CLK_IBUF_BUF_inst/O

net (fo=44, routed)  1.445  14.786  L2/CLK_IBUF_BUF

SLICE_X30Y43     FDRE              r L2/Q_reg[6]/C

clock pessimism    0.000  14.786

clock uncertainty   -0.035  14.751

SLICE_X30Y43     FDRE (Setup_fdre_C_R)  -0.524  14.227  L2/Q_reg[6]
-----
```

```

required time      14.227

arrival time        -10.019
-----
```

```

slack              4.209
```

Slack (MET) : 4.209ns (required time - arrival time)

Source: RESET

(input port clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: L2/Q_reg[7]/R

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Setup (Max at Slow Process Corner)

Requirement: 10.000ns (CLK rise@10.000ns - CLK rise@0.000ns)

Data Path Delay: 10.019ns (logic 1.565ns (15.624%) route 8.453ns (84.376%))

Logic Levels: 2 (IBUF=1 LUT2=1)

Input Delay: 0.000ns

Clock Path Skew: 4.786ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): 4.786ns = (14.786 - 10.000)

Source Clock Delay (SCD): 0.000ns

Clock Pessimism Removal (CPR): 0.000ns

Clock Uncertainty: 0.035ns $((TSJ^2 + TIJ^2)^{1/2} + DJ) / 2 + PE$

Total System Jitter (TSJ): 0.071ns

Total Input Jitter (TIJ): 0.000ns

Discrete Jitter (DJ): 0.000ns

Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
	input delay	0.000	0.000	
U18		0.000	0.000	r RESET (IN)
	net (fo=0)	0.000	0.000	RESET
U18			r	RESET_IBUF_inst/I
U18	IBUF (Prop_ibuf_i_O)	1.441	1.441	r RESET_IBUF_inst/O
	net (fo=14, routed)	6.948	8.389	L2/RESET_IBUF
SLICE_X28Y45			r	L2/Q[0]_i_1/I0
SLICE_X28Y45	LUT2 (Prop_lut2_i0_O)	0.124	8.513	r L2/Q[0]_i_1/O
	net (fo=27, routed)	1.505	10.019	L2/Q[0]_i_1_n_0
SLICE_X30Y43	FDRE		r	L2/Q_reg[7]/R
	(clock CLK rise edge)	10.000	10.000	r
W5		0.000	10.000	r CLK (IN)
	net (fo=0)	0.000	10.000	CLK
W5			r	CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_i_O)	1.388	11.388	r CLK_IBUF_inst/O
	net (fo=1, routed)	1.862	13.250	CLK_IBUF
BUFCTRL_X0Y0			r	CLK_IBUF_BUF_inst/I
BUFCTRL_X0Y0	BUF (Prop_bufg_i_O)	0.091	13.341	r CLK_IBUF_BUF_inst/O
	net (fo=44, routed)	1.445	14.786	L2/CLK_IBUF_BUF
SLICE_X30Y43	FDRE		r	L2/Q_reg[7]/C
	clock pessimism	0.000	14.786	
	clock uncertainty	-0.035	14.751	
SLICE_X30Y43	FDRE (Setup_fdre_C_R)	-0.524	14.227	L2/Q_reg[7]
	required time		14.227	
	arrival time		-10.019	
	slack		4.209	

Min Delay Paths

Slack (MET) : 0.191ns (arrival time - required time)

Source: L0/LIGHT_reg[1]/C

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: L0/LIGHT_reg[1]_replica/D

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (CLK rise@0.000ns - CLK rise@0.000ns)

Data Path Delay: 0.295ns (logic 0.186ns (63.037%) route 0.109ns (36.963%))

Logic Levels: 1 (LUT6=1)

Clock Path Skew: 0.013ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.993ns

Source Clock Delay (SCD): 1.478ns

Clock Pessimism Removal (CPR): 0.502ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5			r	CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_i_O)	0.226	0.226	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.631	0.858	CLK_IBUF
BUFGCTRL_X0Y0			r	CLK_IBUF_BUF inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_i_O)	0.026	0.884	r CLK_IBUF_BUF inst/O
	net (fo=44, routed)	0.595	1.478	L0/CLK_IBUF_BUF
SLICE_X1Y43	FDRE		r	L0/LIGHT_reg[1]/C
SLICE_X1Y43	FDRE (Prop_fdre_c_Q)	0.141	1.619	r L0/LIGHT_reg[1]/Q
	net (fo=7, routed)	0.109	1.728	L0/LIGHT_OBUF[1]
SLICE_X0Y43			r	L0/LIGHT[1]_i_1/I0
SLICE_X0Y43	LUT6 (Prop_lut6_i0_O)	0.045	1.773	r L0/LIGHT[1]_i_1/O
	net (fo=2, routed)	0.000	1.773	L0/LIGHT[1]_i_1_n_0
SLICE_X0Y43	FDRE		r	L0/LIGHT_reg[1]_replica/D
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5			r	CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_i_O)	0.414	0.414	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.685	1.099	CLK_IBUF
BUFGCTRL_X0Y0			r	CLK_IBUF_BUF inst/I

BUFGCTRL_X0Y0	BUFG (Prop_bufg_i_O)	0.029	1.128	r	CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.866	1.993	L0/CLK_IBUF_BUFG	
SLICE_X0Y43	FDRE			r	L0/LIGHT_reg[1]_replica/C
	clock pessimism	-0.502	1.491		
SLICE_X0Y43	FDRE (Hold_fdre_C_D)	0.091	1.582	L0/LIGHT_reg[1]_replica	

	required time		-1.582		
	arrival time		1.773		

	slack		0.191		

Slack (MET) : 0.196ns (arrival time - required time)

Source: L1/Q_2Seconds_reg[1]/C

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: L1/TC2_reg/D

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (CLK rise@0.000ns - CLK rise@0.000ns)

Data Path Delay: 0.317ns (logic 0.246ns (77.542%) route 0.071ns (22.458%))

Logic Levels: 1 (LUT6=1)

Clock Path Skew: 0.000ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.963ns

Source Clock Delay (SCD): 1.449ns

Clock Pessimism Removal (CPR): 0.514ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)

	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_i_O)	0.226	0.226	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.631	0.858	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_i_O)	0.026	0.884	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.566	1.449	L1/CLK_IBUF_BUFG
SLICE_X8Y44	FDRE			r L1/Q_2Seconds_reg[1]/C

SLICE_X8Y44	FDRE (Prop_fdre_C_Q)	0.148	1.597	r L1/Q_2Seconds_reg[1]/Q
	net (fo=3, routed)	0.071	1.668	L1/Q_2Seconds[1]
SLICE_X8Y44				r L1/TC2_i_1/I3
SLICE_X8Y44	LUT6 (Prop_lut6_i3_O)	0.098	1.766	r L1/TC2_i_1/O
	net (fo=1, routed)	0.000	1.766	L1/TC2_i_1_n_0
SLICE_X8Y44	FDRE			r L1/TC2_reg/D

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      (clock CLK rise edge)    0.000  0.000 r

W5              0.000  0.000 r CLK (IN)

      net (fo=0)              0.000  0.000  CLK

W5              r CLK_IBUF_inst/I

W5      IBUF (Prop_ibuf_I_O)    0.414  0.414 r CLK_IBUF_inst/O

      net (fo=1, routed)      0.685  1.099  CLK_IBUF

BUFGCTRL_X0Y0              r CLK_IBUF_BUFG_inst/I

BUFGCTRL_X0Y0  BUFG (Prop_bufg_I_O)  0.029  1.128 r CLK_IBUF_BUFG_inst/O

      net (fo=44, routed)    0.836  1.963  L1/CLK_IBUF_BUFG

SLICE_X8Y44  FDRE              r L1/TC2_reg/C

      clock pessimism        -0.514  1.449

SLICE_X8Y44  FDRE (Hold_fdre_C_D)  0.121  1.570  L1/TC2_reg

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      required time            -1.570

      arrival time             1.766

-----

      slack                    0.196

Slack (MET) :      0.265ns (arrival time - required time)

Source:      L2/Q_reg[14]/C

      (rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination:  L2/Q_reg[14]/D

      (rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group:   CLK

Path Type:    Hold (Min at Fast Process Corner)

Requirement:  0.000ns (CLK rise@0.000ns - CLK rise@0.000ns)

Data Path Delay:  0.399ns (logic 0.274ns (68.589%) route 0.125ns (31.411%))

Logic Levels:  1 (CARRY4=1)

Clock Path Skew:  0.000ns (DCD - SCD - CPR)

Destination Clock Delay (DCD):  1.959ns

Source Clock Delay (SCD):  1.446ns

Clock Pessimism Removal (CPR):  0.513ns
```

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)

	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.226	0.226	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.631	0.858	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026	0.884	r CLK_IBUF_BUFG_inst/O

	net (fo=44, routed)	0.563	1.446	L2/CLK_IBUF_BUF
SLICE_X30Y45	FDRE			r L2/Q_reg[14]/C

SLICE_X30Y45	FDRE (Prop_fdre_C_Q)	0.164	1.610	r L2/Q_reg[14]/Q
	net (fo=3, routed)	0.125	1.736	L2/Q_reg[14]
SLICE_X30Y45				r L2/Q_reg[12]_i_1/S[2]
SLICE_X30Y45	CARRY4 (Prop_carry4_S[2]_O[2])			
		0.110	1.846	r L2/Q_reg[12]_i_1/O[2]
	net (fo=1, routed)	0.000	1.846	L2/Q_reg[12]_i_1_n_5
SLICE_X30Y45	FDRE			r L2/Q_reg[14]/D

	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.414	0.414	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.685	1.099	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUF_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.029	1.128	r CLK_IBUF_BUF_inst/O
	net (fo=44, routed)	0.832	1.959	L2/CLK_IBUF_BUF
SLICE_X30Y45	FDRE			r L2/Q_reg[14]/C
	clock pessimism	-0.513	1.446	
SLICE_X30Y45	FDRE (Hold_fdre_C_D)	0.134	1.580	L2/Q_reg[14]

	required time		-1.580	
	arrival time		1.846	

	slack		0.265	
Slack (MET) :	0.267ns (arrival time - required time)			
Source:	L2/Q_reg[10]/C			
	(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})			
Destination:	L2/Q_reg[10]/D			
	(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})			
Path Group:	CLK			
Path Type:	Hold (Min at Fast Process Corner)			
Requirement:	0.000ns (CLK rise@0.000ns - CLK rise@0.000ns)			
Data Path Delay:	0.401ns (logic 0.274ns (68.412%) route 0.127ns (31.588%))			
Logic Levels:	1 (CARRY4=1)			
Clock Path Skew:	0.000ns (DCD - SCD - CPR)			
Destination Clock Delay (DCD):	1.959ns			
Source Clock Delay (SCD):	1.446ns			
Clock Pessimism Removal (CPR):	0.513ns			

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.226	0.226	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.631	0.858	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026	0.884	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.563	1.446	L2/CLK_IBUF_BUFG
SLICE_X30Y44	FDRE			r L2/Q_reg[10]/C
SLICE_X30Y44	FDRE (Prop_fdre_C_Q)	0.164	1.610	r L2/Q_reg[10]/Q
	net (fo=2, routed)	0.127	1.737	L2/Q_reg[10]
SLICE_X30Y44				r L2/Q_reg[8]_I_1/S[2]
SLICE_X30Y44	CARRY4 (Prop_carry4_S[2]_O[2])			
		0.110	1.847	r L2/Q_reg[8]_I_1/O[2]
	net (fo=1, routed)	0.000	1.847	L2/Q_reg[8]_I_1_n_5
SLICE_X30Y44	FDRE			r L2/Q_reg[10]/D
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.414	0.414	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.685	1.099	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.029	1.128	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.832	1.959	L2/CLK_IBUF_BUFG
SLICE_X30Y44	FDRE			r L2/Q_reg[10]/C
	clock pessimism	-0.513	1.446	
SLICE_X30Y44	FDRE (Hold_fdre_C_D)	0.134	1.580	L2/Q_reg[10]
	required time		-1.580	
	arrival time		1.847	
	slack		0.267	
Slack (MET) :	0.267ns (arrival time - required time)			
Source:	L2/Q_reg[18]/C			
	(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})			
Destination:	L2/Q_reg[18]/D			
	(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})			

Path Group: CLK

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (CLK rise@0.000ns - CLK rise@0.000ns)

Data Path Delay: 0.401ns (logic 0.274ns (68.412%) route 0.127ns (31.588%))

Logic Levels: 1 (CARRY4=1)

Clock Path Skew: 0.000ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.959ns

Source Clock Delay (SCD): 1.446ns

Clock Pessimism Removal (CPR): 0.513ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.226	0.226	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.631	0.858	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026	0.884	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.563	1.446	L2/CLK_IBUF_BUFG
SLICE_X30Y46	FDRE			r L2/Q_reg[18]/C
SLICE_X30Y46	FDRE (Prop_fdre_C_Q)	0.164	1.610	r L2/Q_reg[18]/Q
	net (fo=3, routed)	0.127	1.737	L2/Q_reg[18]
SLICE_X30Y46				r L2/Q_reg[16]_i_1/S[2]
SLICE_X30Y46	CARRY4 (Prop_carry4_S[2]_O[2])			
		0.110	1.847	r L2/Q_reg[16]_i_1/O[2]
	net (fo=1, routed)	0.000	1.847	L2/Q_reg[16]_i_1_n_5
SLICE_X30Y46	FDRE			r L2/Q_reg[18]/D
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.414	0.414	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.685	1.099	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.029	1.128	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.832	1.959	L2/CLK_IBUF_BUFG
SLICE_X30Y46	FDRE			r L2/Q_reg[18]/C
	clock pessimism	-0.513	1.446	
SLICE_X30Y46	FDRE (Hold_fdre_C_D)	0.134	1.580	L2/Q_reg[18]

required time	-1.580
arrival time	1.847

slack	0.267
-------	-------

Slack (MET) : 0.267ns (arrival time - required time)

Source: L2/Q_reg[2]/C
(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: L2/Q_reg[2]/D
(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (CLK rise@0.000ns - CLK rise@0.000ns)

Data Path Delay: 0.401ns (logic 0.274ns (68.412%) route 0.127ns (31.588%))

Logic Levels: 1 (CARRY4=1)

Clock Path Skew: 0.000ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.958ns

Source Clock Delay (SCD): 1.445ns

Clock Pessimism Removal (CPR): 0.513ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5			r	CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_i_O)	0.226	0.226	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.631	0.858	CLK_IBUF
BUFCTRL_X0Y0			r	CLK_IBUF_BUF_inst/I
BUFCTRL_X0Y0	BUF (Prop_bufg_i_O)	0.026	0.884	r CLK_IBUF_BUF_inst/O
	net (fo=44, routed)	0.562	1.445	L2/CLK_IBUF_BUF
SLICE_X30Y42	FDRE		r	L2/Q_reg[2]/C
SLICE_X30Y42	FDRE (Prop_fdre_C_Q)	0.164	1.609	r L2/Q_reg[2]/Q
	net (fo=3, routed)	0.127	1.736	L2/Q_reg[2]
SLICE_X30Y42			r	L2/Q_reg[0]_i_2/S[2]
SLICE_X30Y42	CARRY4 (Prop_carry4_S[2]_O[2])			
		0.110	1.846	r L2/Q_reg[0]_i_2/O[2]
	net (fo=1, routed)	0.000	1.846	L2/Q_reg[0]_i_2_n_5
SLICE_X30Y42	FDRE		r	L2/Q_reg[2]/D
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK

W5				r	CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.414	0.414	r	CLK_IBUF_inst/O
	net (fo=1, routed)	0.685	1.099	CLK_IBUF	
BUFGCTRL_X0Y0				r	CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.029	1.128	r	CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.831	1.958	L2/CLK_IBUF_BUFG	
SLICE_X30Y42	FDRE			r	L2/Q_reg[2]/C
	clock pessimism	-0.513	1.445		
SLICE_X30Y42	FDRE (Hold_fdre_C_D)	0.134	1.579	L2/Q_reg[2]	

	required time	-1.579
	arrival time	1.846
	slack	0.267

Slack (MET) : 0.267ns (arrival time - required time)

Source: L2/Q_reg[6]/C

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: L2/Q_reg[6]/D

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (CLK rise@0.000ns - CLK rise@0.000ns)

Data Path Delay: 0.401ns (logic 0.274ns (68.412%) route 0.127ns (31.588%))

Logic Levels: 1 (CARRY4=1)

Clock Path Skew: 0.000ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.959ns

Source Clock Delay (SCD): 1.446ns

Clock Pessimism Removal (CPR): 0.513ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.226	0.226	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.631	0.858	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026	0.884	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.563	1.446	L2/CLK_IBUF_BUFG
SLICE_X30Y43	FDRE			r L2/Q_reg[6]/C
SLICE_X30Y43	FDRE (Prop_fdre_C_Q)	0.164	1.610	r L2/Q_reg[6]/Q
	net (fo=3, routed)	0.127	1.737	L2/Q_reg[6]

W5		r CLK_IBUF_inst/I		
W5	IBUF (Prop_ibuf_I_O)	0.226	0.226	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.631	0.858	CLK_IBUF
BUFGCTRL_X0Y0		r CLK_IBUF_BUFG_inst/I		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026	0.884	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.564	1.447	L2/CLK_IBUF_BUFG
SLICE_X30Y47	FDRE			r L2/Q_reg[22]/C

SLICE_X30Y47	FDRE (Prop_fdre_C_Q)	0.164	1.611	r L2/Q_reg[22]/Q
	net (fo=3, routed)	0.127	1.738	L2/Q_reg[22]
SLICE_X30Y47				r L2/Q_reg[20]_i_1/S[2]
SLICE_X30Y47	CARRY4 (Prop_carry4_S[2]_O[2])			
		0.110	1.848	r L2/Q_reg[20]_i_1/O[2]
	net (fo=1, routed)	0.000	1.848	L2/Q_reg[20]_i_1_n_5
SLICE_X30Y47	FDRE			r L2/Q_reg[22]/D

	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.414	0.414	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.685	1.099	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.029	1.128	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.833	1.960	L2/CLK_IBUF_BUFG
SLICE_X30Y47	FDRE			r L2/Q_reg[22]/C
	clock pessimism	-0.513	1.447	
SLICE_X30Y47	FDRE (Hold_fdre_C_D)	0.134	1.581	L2/Q_reg[22]

	required time	-1.581
	arrival time	1.848

	slack	0.267
--	-------	-------

Slack (MET) : 0.267ns (arrival time - required time)

Source: L2/Q_reg[26]/C

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: L2/Q_reg[26]/D

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (CLK rise@0.000ns - CLK rise@0.000ns)

Data Path Delay: 0.401ns (logic 0.274ns (68.412%) route 0.127ns (31.588%))

Logic Levels: 1 (CARRY4=1)

Clock Path Skew: 0.000ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.960ns

Source Clock Delay (SCD): 1.447ns

Clock Pessimism Removal (CPR): 0.513ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.226	0.226	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.631	0.858	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026	0.884	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.564	1.447	L2/CLK_IBUF_BUFG
SLICE_X30Y48	FDRE			r L2/Q_reg[26]/C
SLICE_X30Y48	FDRE (Prop_fdre_C_Q)	0.164	1.611	r L2/Q_reg[26]/Q
	net (fo=3, routed)	0.127	1.738	L2/Q_reg[26]
SLICE_X30Y48				r L2/Q_reg[24]_i_1/S[2]
SLICE_X30Y48	CARRY4 (Prop_carry4_S[2]_O[2])			
		0.110	1.848	r L2/Q_reg[24]_i_1/O[2]
	net (fo=1, routed)	0.000	1.848	L2/Q_reg[24]_i_1_n_5
SLICE_X30Y48	FDRE			r L2/Q_reg[26]/D
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5				r CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_I_O)	0.414	0.414	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.685	1.099	CLK_IBUF
BUFGCTRL_X0Y0				r CLK_IBUF_BUFG_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.029	1.128	r CLK_IBUF_BUFG_inst/O
	net (fo=44, routed)	0.833	1.960	L2/CLK_IBUF_BUFG
SLICE_X30Y48	FDRE			r L2/Q_reg[26]/C
	clock pessimism	-0.513	1.447	
SLICE_X30Y48	FDRE (Hold_fdre_C_D)	0.134	1.581	L2/Q_reg[26]
	required time	-1.581		
	arrival time	1.848		
	slack	0.267		

Slack (MET) : 0.275ns (arrival time - required time)

Source: L0/FSM_sequential_STATE_reg[1]/C

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: L0/FSM_sequential_STATE_reg[1]/D

(rising edge-triggered cell FDRE clocked by CLK {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: CLK

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (CLK rise@0.000ns - CLK rise@0.000ns)

Data Path Delay: 0.366ns (logic 0.186ns (50.768%) route 0.180ns (49.232%))

Logic Levels: 1 (LUT5=1)

Clock Path Skew: 0.000ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.959ns

Source Clock Delay (SCD): 1.446ns

Clock Pessimism Removal (CPR): 0.513ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5			r	CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_i_O)	0.226	0.226	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.631	0.858	CLK_IBUF
BUFGCTRL_X0Y0			r	CLK_IBUF_BUF_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_i_O)	0.026	0.884	r CLK_IBUF_BUF_inst/O
	net (fo=44, routed)	0.563	1.446	L0/CLK_IBUF_BUF
SLICE_X33Y43	FDRE		r	L0/FSM_sequential_STATE_reg[1]/C
SLICE_X33Y43	FDRE (Prop_fdre_C_Q)	0.141	1.587	r L0/FSM_sequential_STATE_reg[1]/Q
	net (fo=5, routed)	0.180	1.768	L0/STATE[1]
SLICE_X33Y43			r	L0/FSM_sequential_STATE[1]_i_1/I0
SLICE_X33Y43	LUT5 (Prop_lut5_i0_O)	0.045	1.813	r L0/FSM_sequential_STATE[1]_i_1/O
	net (fo=1, routed)	0.000	1.813	L0/FSM_sequential_STATE[1]_i_1_n_0
SLICE_X33Y43	FDRE		r	L0/FSM_sequential_STATE_reg[1]/D
	(clock CLK rise edge)	0.000	0.000	r
W5		0.000	0.000	r CLK (IN)
	net (fo=0)	0.000	0.000	CLK
W5			r	CLK_IBUF_inst/I
W5	IBUF (Prop_ibuf_i_O)	0.414	0.414	r CLK_IBUF_inst/O
	net (fo=1, routed)	0.685	1.099	CLK_IBUF
BUFGCTRL_X0Y0			r	CLK_IBUF_BUF_inst/I
BUFGCTRL_X0Y0	BUFG (Prop_bufg_i_O)	0.029	1.128	r CLK_IBUF_BUF_inst/O
	net (fo=44, routed)	0.832	1.959	L0/CLK_IBUF_BUF

SLICE_X33Y43	FDRE	r	L0/FSM_sequential_STATE_reg[1]/C
clock pessimism	-0.513	1.446	
SLICE_X33Y43	FDRE (Hold_fdre_C_D)	0.091	1.537 L0/FSM_sequential_STATE_reg[1]
required time	-1.537		
arrival time	1.813		
slack	0.275		

Pulse Width Checks

Clock Name: CLK

Waveform(ns): { 0.000 5.000 }

Period(ns): 10.000

Sources: { CLK }

Check Type	Corner	Lib Pin	Reference Pin	Required(ns)	Actual(ns)	Slack(ns)	Location	Pin
Min Period	n/a	BUFG/I	n/a	2.155	10.000	7.845	BUFGCTRL_X0Y0 CLK_IBUF_BUFG_inst/I	
Min Period	n/a	FDRE/C	n/a	1.000	10.000	9.000	SLICE_X33Y43 L0/FSM_sequential_STATE_reg[1]/C	
Min Period	n/a	FDRE/C	n/a	1.000	10.000	9.000	SLICE_X1Y3 L0/LIGHT_reg[0]/C	
Min Period	n/a	FDRE/C	n/a	1.000	10.000	9.000	SLICE_X1Y43 L0/LIGHT_reg[1]/C	
Min Period	n/a	FDSE/C	n/a	1.000	10.000	9.000	SLICE_X1Y20 L0/LIGHT_reg[2]/C	
Min Period	n/a	FDSE/C	n/a	1.000	10.000	9.000	SLICE_X65Y61 L0/LIGHT_reg[3]/C	
Min Period	n/a	FDRE/C	n/a	1.000	10.000	9.000	SLICE_X65Y87 L0/LIGHT_reg[4]/C	
Min Period	n/a	FDRE/C	n/a	1.000	10.000	9.000	SLICE_X33Y43 L0/SEL_reg/C	
Min Period	n/a	FDRE/C	n/a	1.000	10.000	9.000	SLICE_X28Y45 L1/Q_10Seconds_reg[0]/C	
Min Period	n/a	FDRE/C	n/a	1.000	10.000	9.000	SLICE_X28Y45 L1/Q_10Seconds_reg[1]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X33Y43 L0/FSM_sequential_STATE_reg[1]/C	
Low Pulse Width	Slow	FDSE/C	n/a	0.500	5.000	4.500	SLICE_X65Y61 L0/LIGHT_reg[3]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X65Y87 L0/LIGHT_reg[4]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X33Y43 L0/SEL_reg/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X28Y45 L1/Q_10Seconds_reg[0]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X28Y45 L1/Q_10Seconds_reg[1]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X28Y45 L1/Q_10Seconds_reg[2]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X28Y45 L1/Q_10Seconds_reg[3]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X28Y44 L1/TC10_reg/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X30Y44 L2/Q_reg[10]/C	
High Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X1Y3 L0/LIGHT_reg[0]/C	
High Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X1Y43 L0/LIGHT_reg[1]/C	
High Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X8Y44 L1/Q_2Seconds_reg[0]/C	
High Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X8Y44 L1/Q_2Seconds_reg[1]/C	

High Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X8Y44	L1/TC2_reg/C
High Pulse Width	Slow	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X0Y43	L0/LIGHT_reg[1]_replica/C
High Pulse Width	Fast	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X33Y43	L0/FSM_sequential_STATE_reg[1]/C
High Pulse Width	Fast	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X1Y3	L0/LIGHT_reg[0]/C
High Pulse Width	Fast	FDRE/C	n/a	0.500	5.000	4.500	SLICE_X1Y43	L0/LIGHT_reg[1]/C
High Pulse Width	Fast	FDSE/C	n/a	0.500	5.000	4.500	SLICE_X1Y20	L0/LIGHT_reg[2]/C

| Data sheet

Input Ports Setup/Hold

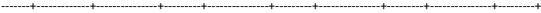
Reference	Input	IO Reg	Delay	Setup(ns)	Process	Hold(ns)	Process	Internal
Clock	Port	Type	Type	to Clk (Edge)	Corner	to Clk (Edge)	Corner	Clock
CLK	RESET	FDRE	-	6.581 (r)	SLOW	-0.328 (r)	FAST	

Output Ports Clock-to-out

Reference	Output	IO Reg	Delay	Max Clk (Edge)	Process	Min Clk (Edge)	Process	Internal
Clock	Port	Type	Type	to port(ns)	Corner	to port(ns)	Corner	Clock
CLK	LIGHT[0]	FDRE	-	10.685 (r)	SLOW	3.072 (r)	FAST	
CLK	LIGHT[1]	FDRE	-	10.713 (r)	SLOW	3.100 (r)	FAST	
CLK	LIGHT[2]	FDSE	-	10.668 (r)	SLOW	3.059 (r)	FAST	
CLK	LIGHT[3]	FDSE	-	10.676 (r)	SLOW	3.077 (r)	FAST	
CLK	LIGHT[4]	FDRE	-	10.681 (r)	SLOW	3.082 (r)	FAST	

Setup between Clocks

Source	Destination	Src:Rise	Process	Src:Rise	Process	Src:Fall	Process	Src:Fall	Process
Clock	Clock	Dest:Rise(ns)	Corner	Dest:Fall(ns)	Corner	Dest:Rise(ns)	Corner	Dest:Fall(ns)	Corner
CLK	CLK	5.755	SLOW						



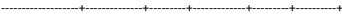
Max / Min delays for output bus

Clocked by: CLK

Bus Skew: 0.045 ns



		Max	Process	Min	Process	Edge	
Pad		Delay(ns)	Corner	Delay(ns)	Corner	Skew(ns)	



LIGHT[0]		10.685 (r)	SLOW	3.072 (r)	FAST	0.017	
LIGHT[1]		10.713 (r)	SLOW	3.100 (r)	FAST	0.045	
LIGHT[2]		10.668 (r)	SLOW	3.059 (r)	FAST	0.000	
LIGHT[3]		10.676 (r)	SLOW	3.077 (r)	FAST	0.017	
LIGHT[4]		10.681 (r)	SLOW	3.082 (r)	FAST	0.022	



Worst Case Summary | 10.713 (r) | SLOW | 3.059 (r) | FAST | 0.045 |



AREA:

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| Tool Version : Vivado v.2020.1 (win64) Build 2902540 Wed May 27 19:54:49 MDT 2020

| Date : Wed Aug 6 19:00:24 2025

| Host : ELEC-403-103 running 64-bit major release (build 9200)

| Command : report_utilization -file Z:/VHDL2/lab_28/project_1/utilization_report.txt -name utilization_3

| Design : Top_level

| Device : 7a35tcpg236-1

| Design State : Routed



Utilization Design Information

Table of Contents



- 1. Slice Logic
 - 1.1 Summary of Registers by Type
- 2. Slice Logic Distribution
- 3. Memory
- 4. DSP
- 5. IO and GT Specific
- 6. Clocking
- 7. Specific Feature
- 8. Primitives

9. Black Boxes

10. Instantiated Netlists

1. Slice Logic

	Site Type		Used Fixed	Available	Util%
	Slice LUTs		29 0	20800	0.14
	LUT as Logic		29 0	20800	0.14
	LUT as Memory		0 0	9600	0.00
	Slice Registers		44 0	41600	0.11
	Register as Flip Flop		44 0	41600	0.11
	Register as Latch		0 0	41600	0.00
	F7 Muxes		0 0	16300	0.00
	F8 Muxes		0 0	8150	0.00

1.1 Summary of Registers by Type

	Total		Clock Enable Synchronous	Asynchronous	
	0		_ -	-	
	0		_ -	Set	
	0		_ -	Reset	
	0		_ Set	-	
	0		_ Reset	-	
	0		Yes -	-	
	0		Yes -	Set	
	0		Yes -	Reset	
	2		Yes Set	-	
	42		Yes Reset	-	

2. Slice Logic Distribution

	Site Type		Used Fixed	Available	Util%

	Site Type	Used	Fixed	Available	Util%
Bonded IOB		7	7	106	6.60
IOB Master Pads		2			
IOB Slave Pads		5			
Bonded IPADs		0	0	10	0.00
Bonded OPADs		0	0	4	0.00
PHY_CONTROL		0	0	5	0.00
PHASER_REF		0	0	5	0.00
OUT_FIFO		0	0	20	0.00
IN_FIFO		0	0	20	0.00
IDELAYCTRL		0	0	5	0.00
IBUFDS		0	0	104	0.00
GTPE2_CHANNEL		0	0	2	0.00
PHASER_OUT/PHASER_OUT_PHY		0	0	20	0.00
PHASER_IN/PHASER_IN_PHY		0	0	20	0.00
IDELAYE2/IDELAYE2_FINEDELAY		0	0	250	0.00
IBUFDS_GTE2		0	0	2	0.00
ILOGIC		0	0	106	0.00
OLOGIC		0	0	106	0.00

6. Clocking

	Site Type	Used	Fixed	Available	Util%
BUFGCTRL		1	0	32	3.13
BUFIO		0	0	20	0.00
MMCM2_ADV		0	0	5	0.00
PLLE2_ADV		0	0	5	0.00
BUFMRC		0	0	10	0.00
BUFHCE		0	0	72	0.00
BUFR		0	0	20	0.00

7. Specific Feature

	Site Type	Used	Fixed	Available	Util%

Site Type	Used	Fixed	Available	Util%
BSCANE2	0	0	4	0.00
CAPTUREE2	0	0	1	0.00
DNA_PORT	0	0	1	0.00
EFUSE_USR	0	0	1	0.00
FRAME_ECCE2	0	0	1	0.00
ICAPE2	0	0	2	0.00
PCIE_2_1	0	0	1	0.00
STARTUPE2	0	0	1	0.00
XADC	0	0	1	0.00

8. Primitives

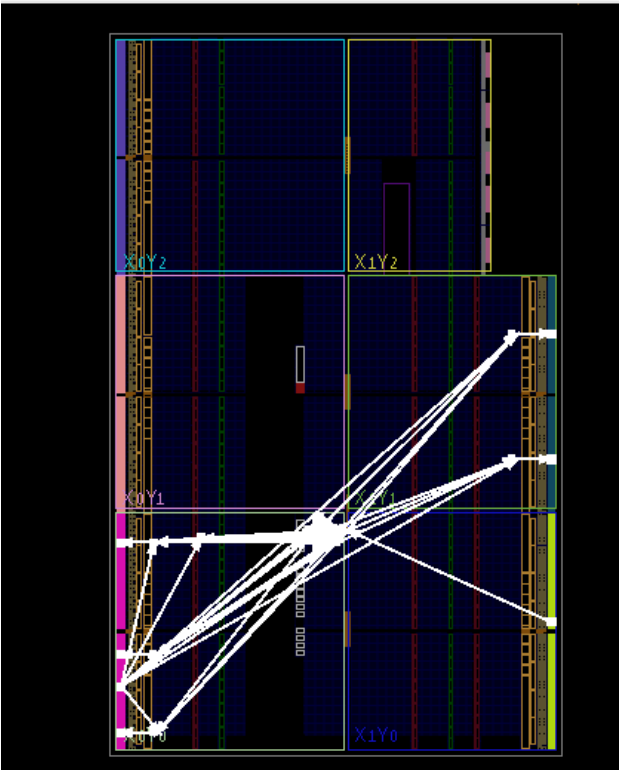
Ref Name	Used	Functional Category
FDRE	42	Flop & Latch
LUT2	26	LUT
CARRY4	11	CarryLogic
LUT5	9	LUT
OBUF	5	IO
LUT3	4	LUT
LUT1	4	LUT
LUT6	2	LUT
IBUF	2	IO
FDSE	2	Flop & Latch
LUT4	1	LUT
BUFG	1	Clock

9. Black Boxes

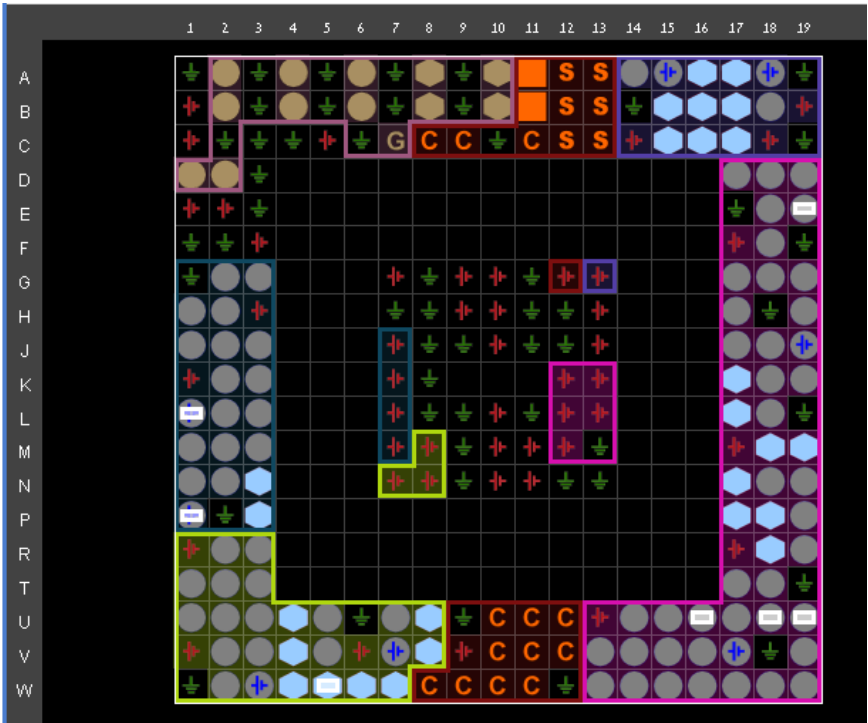
Ref Name	Used

10. Instantiated Netlists

DEVICE:



I/O:



סימולציה לאחר

P&R:

